An Introduction to IoT Protocols

Across the globe, billions of devices are today communicating and exchanging data with each other, thanks to the Internet of Things (IoT). IoT communication protocols protect and ensure security to the data being exchanged between these devices. This article introduces the most popular protocols in use today.

Things around us are not only getting smarter and intelligent, but also connecting and interacting seamlessly with each other. The Internet of Things (IoT) refers to a collection of devices that are connected to the Internet, embedded with electronics, software, sensors, actuators, and network connectivity enabling these objects to collect and exchange data. The data gathered from IoT can be used to predict, learn, and make real-time decisions. IoT plays a significant role in applications like healthcare, smart homes, transportation, automation, agriculture, vehicles, and disaster recovery.

The IoT system can function and transfer information only when devices are online and safely connected to a communications network. This is where IoT standards and protocols make an entry. IoT devices can be connected either using an IP or a non-IP network. IP network connections are relatively complex and require increased memory and power from IoT devices, although range is not a problem. On the other hand, non-IP networks demand relatively less power and memory but have a range limitation.

IoT protocol architecture

IoT architecture depends on its functionality and implementation in different sectors. The basic process flow, on which IoT is built, has two important architectures -- the 3-layer architecture and 5-layer architecture.

3-layer IoT architecture: The 3-layer architecture is the most basic one. It comprises three layers, namely, the perception, network and application layers.

- The perception layer is the physical layer, which includes all the smart sensor-based devices that collect the data from the environment.
- The network layer includes all the wireless and the wired communication technologies, and is responsible for providing connections between the devices and the applications of the IoT ecosystem. The data is then passed on to the application layer.
- The application layer is accountable for delivering application-specific services to the user. It defines